

11.49

$$(a) H(s) = \frac{Ga}{s+a} = G$$

$$(b) H(s) = \frac{Ga}{s+a} \Leftrightarrow h(t) = Ga e^{-at} \Rightarrow \tau = \frac{1}{a}$$

$$(c) H(j\omega) = \frac{Ga}{j\omega+a}, \quad |j\omega+a| = \sqrt{a^2+\omega^2}$$

$$\left| \frac{Ga}{j\omega+a} \right| = \frac{1}{\sqrt{2}} |G| \Leftrightarrow \sqrt{2} |a| = |j\omega+a| = \sqrt{a^2+\omega^2} \Rightarrow \omega = a.$$

$$\Rightarrow \text{bandwidth} = \frac{\omega}{2\pi} = \frac{a}{2\pi}$$

$$(d) Q(s) = \frac{\frac{Ga}{s+a}}{1+K \cdot \frac{Ga}{s+a}}, \quad Q(0) = \frac{G}{1+KG}$$

$$\doteq \frac{Ga}{s+a(1+KG)}, \quad \tau = \frac{1}{a(1+KG)}$$

$$\text{bandwidth} = \frac{a(1+KG)}{2\pi}$$

$$(e) 1+KG = 2 \Leftrightarrow K = \frac{1}{G}, \quad \tau \text{ 變為開環的一半}$$

bandwidth 變為開環的 2 倍

11.53

(a) According the theorem of Kirchoff, we get.

$$-V_o + V_d + iR + V_i = 0 \Leftrightarrow V_o(t) = V_d(t) + Ri_d(t) + V_i(t)$$

$$V_o(t) = K(0 - (-V_d + V_o(t))) = -K(V_o(t) - V_d(t))$$

$$(b) \cancel{V_o(t) = K(V_d(t) - V_o(t))} = K(V_d(t) - V_o(t))$$

$$\cancel{V_i(t) = \frac{V_o(t)}{K} = RM \exp(\beta V_o(t)/kT)}$$

$$V_d(t) = (1 + \frac{1}{K}) V_o(t)$$

$$V_o(t) = (1 + \frac{1}{K}) V_o(t) + RM \exp(\beta (1 + \frac{1}{K}) V_o(t)/kT) + V_i(t) \quad \left[\text{Figure P11.53(a)} \right]$$

$$\text{When } K \rightarrow \infty, \quad V_i(t) = -RM \exp(\beta V_o/kT)$$

$$V_o(t) = -K(V_i + RM \exp(\beta V_o/kT))$$

[Figure P11.53(b)]

$$\text{When } K \rightarrow \infty, \quad V_i = -RM \exp(\beta V_o/kT).$$

$$(c) \cancel{V_o(t)} = \frac{V_i}{-RM} = \exp(\beta V_o/kT) \Leftrightarrow V_o = \frac{kT}{\beta} \ln\left(-\frac{V_i}{RM}\right)$$